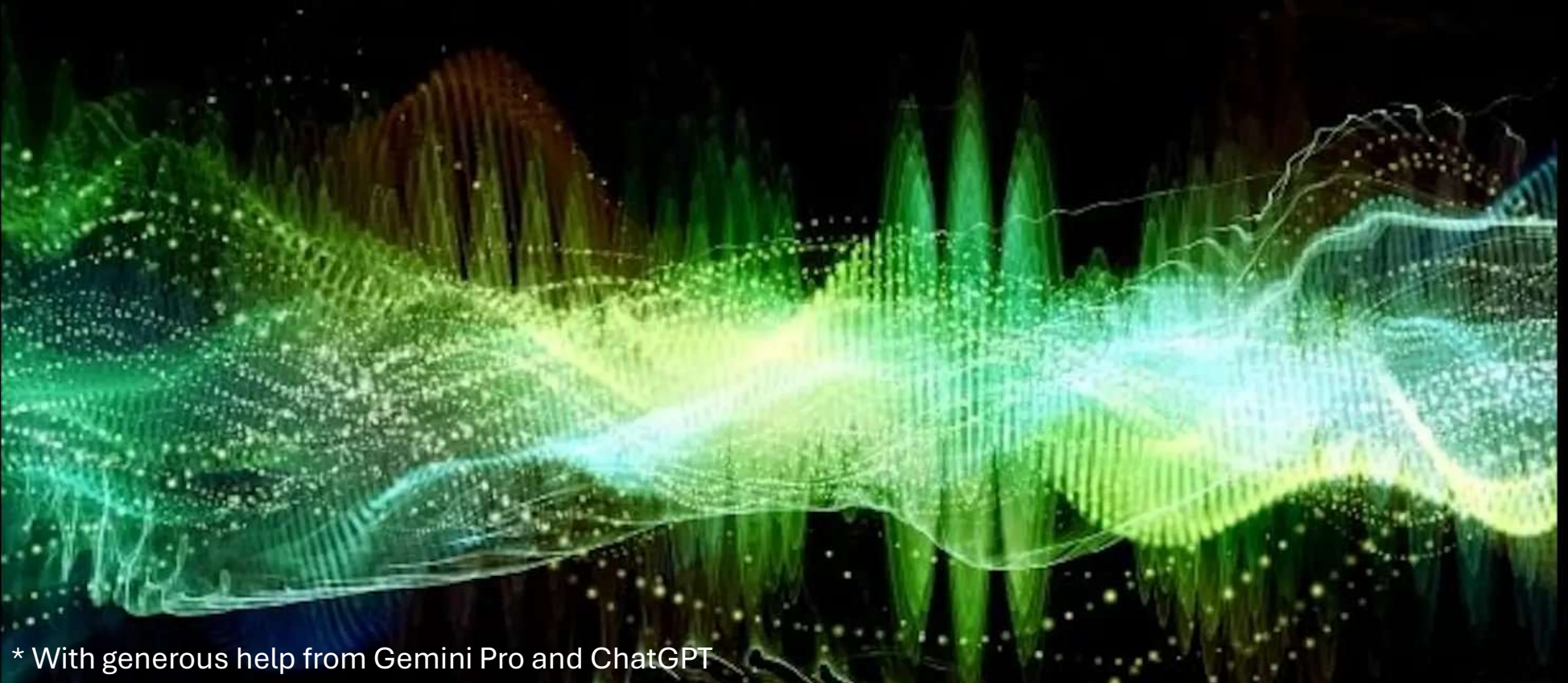


ECEN 4/5632 Theory and Application of Digital Filters

Lecture 10, 9-29-26, 10-01-26

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* With generous help from Gemini Pro and ChatGPT

Linear Phase, Generalized Phase, and ML Intersections

From symmetry constraints to the four FIR types and phase-aware Conv1d kernels

generalized phase → FIR Types I–IV → zero constellations → Conv1d bias

75 minutes • senior + first-year graduate ECE

Equations typeset in standard mathematical notation

Roadmap

- 0–25 min • Derive generalized linear phase and why practical exact linear-phase rational filters are FIR.
- 25–50 min • Use symmetry / antisymmetry to classify FIR Types I–IV and their forced endpoint zeros.
- 50–70 min • Interpret a Conv1d kernel as a trainable FIR and ask how to enforce linear-phase inductive bias.
- 70–75 min • Bridge to multirate filtering: alias suppression needs magnitude selectivity; linear phase is often desirable for alignment.

Generalized linear phase

$$\angle H(e^{j\omega}) = \beta - \alpha\omega$$

constant (pointing to β)
variable in ω (under $\alpha\omega$)
 α : constant

$$H(e^{j\omega}) = e^{-j\alpha\omega} e^{j\beta} A(\omega)$$

- $A(\omega)$ is real for the standard real-coefficient linear-phase FIR cases.
- Its sign can cause π phase jumps without changing the underlying constant group delay.
- The physically important quantity is $\tau_g = \alpha$.

real valued if FIR filter has specific symmetries (with arrow pointing to $A(\omega)$)

Why exact linear phase points to FIR structure

- A stable causal rational IIR has poles whose phase contributions vary with frequency.
- To obtain exact constant group delay over the full usable band, reciprocal pole effects would need exact cancellation; stable causal cancellation reduces to trivial scale/delay behavior in the practical rational setting.
- Thus the nontrivial exact linear-phase filters used in DSP are FIR filters with symmetry constraints.

For exact linear phase we need to be able to break up $H(e^{j\omega})$ into $A(\omega)$ real valued times $e^{j(\beta - \alpha\omega)}$

$\nearrow$
 $A(e^{j\omega})$ DTFT that is real-valued if $A(e^{j\omega}) \Leftrightarrow a[n]$ is real and even

Nuance

Pure delays and trivial scale/delay factors are FIR anyway; “only FIR” should be understood in the standard stable causal rational filter-design context.

Symmetry gives generalized linear phase

type I, II FIR

$$h[n] = h[M - n]$$

Symmetric

Symmetric

Pairing n with $M-n$ yields cosine terms after factoring out $e^{-j\omega M/2}$.

type III, IV FIR

$$h[n] = -h[M - n]$$

Antisymmetric

Antisymmetric

Pairing produces sine terms and an additional $\pm\pi/2$ phase offset.

$$H(e^{j\omega}) = e^{-j\omega M/2} A(\omega)$$

Symmetry gives generalized linear phase

Let $h[n] = h[M-n]$ $0 \leq n \leq M$ order M
 $N = M+1$ taps assume M even

$$H(e^{j\omega}) = \sum_{n=0}^M h[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{M-1} h[n] e^{-j\omega n} + \underbrace{h[M-n]}_{\text{"h[n]"}} e^{-j\omega(M-n)} + h\left[\frac{M}{2}\right] e^{-j\omega \frac{M}{2}}$$

$$= \sum_{n=0}^{M-1} h[n] \left(e^{-j\omega n} + e^{j\omega n} e^{-j\omega M} \right) + h\left[\frac{M}{2}\right] e^{-j\omega \frac{M}{2}}$$

$$\left(e^{-j\omega n} e^{j\omega \frac{M}{2}} + e^{j\omega n} e^{-j\omega \frac{M}{2}} \right) e^{-j\omega \frac{M}{2}}$$

$$= 2 \cos\left(\omega\left(\frac{M}{2} - n\right)\right)$$

$$= \left(\sum_{n=0}^{M-1} 2h[n] \cos\left(\omega\left(\frac{M}{2} - n\right)\right) + h\left[\frac{M}{2}\right] \right) e^{-j\omega \frac{M}{2}}$$

entirely real valued = $A(\omega)$ but can be bipolar

linear phase
 $\tau_g = M/2$ constant

Symmetric derivation by pairing taps

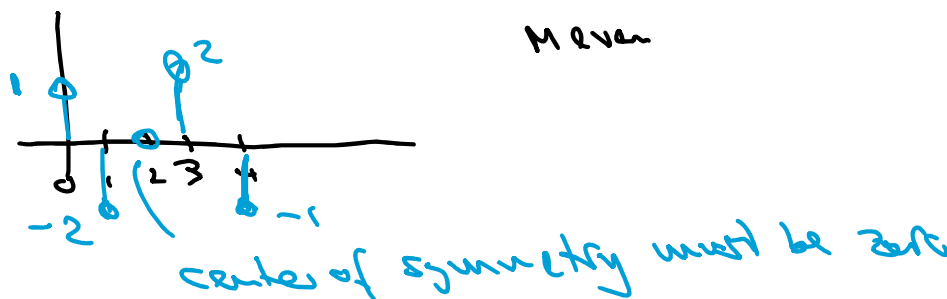
- Start with $H(e^{j\omega}) = \sum_{n=0}^{M-1} h[n] e^{-j\omega n}$.
- Pair n with $M-n$ using $h[n] = h[M-n]$.
- Each pair becomes a real cosine factor times $e^{-j\omega M/2}$.
- Therefore all non-sign phase is the pure delay term $-\omega M/2$.



$$\tau_g = \frac{M}{2} = \frac{N-1}{2}$$

Antisymmetry gives a sine-weighted amplitude

- For $h[n] = -h[M-n]$, paired terms become proportional to $\sin((M/2-n)\omega)$.
- The frequency response is again a pure delay times a real amplitude, with a fixed quadrature factor.
- That symmetry forces endpoint zeros that determine which responses are possible.



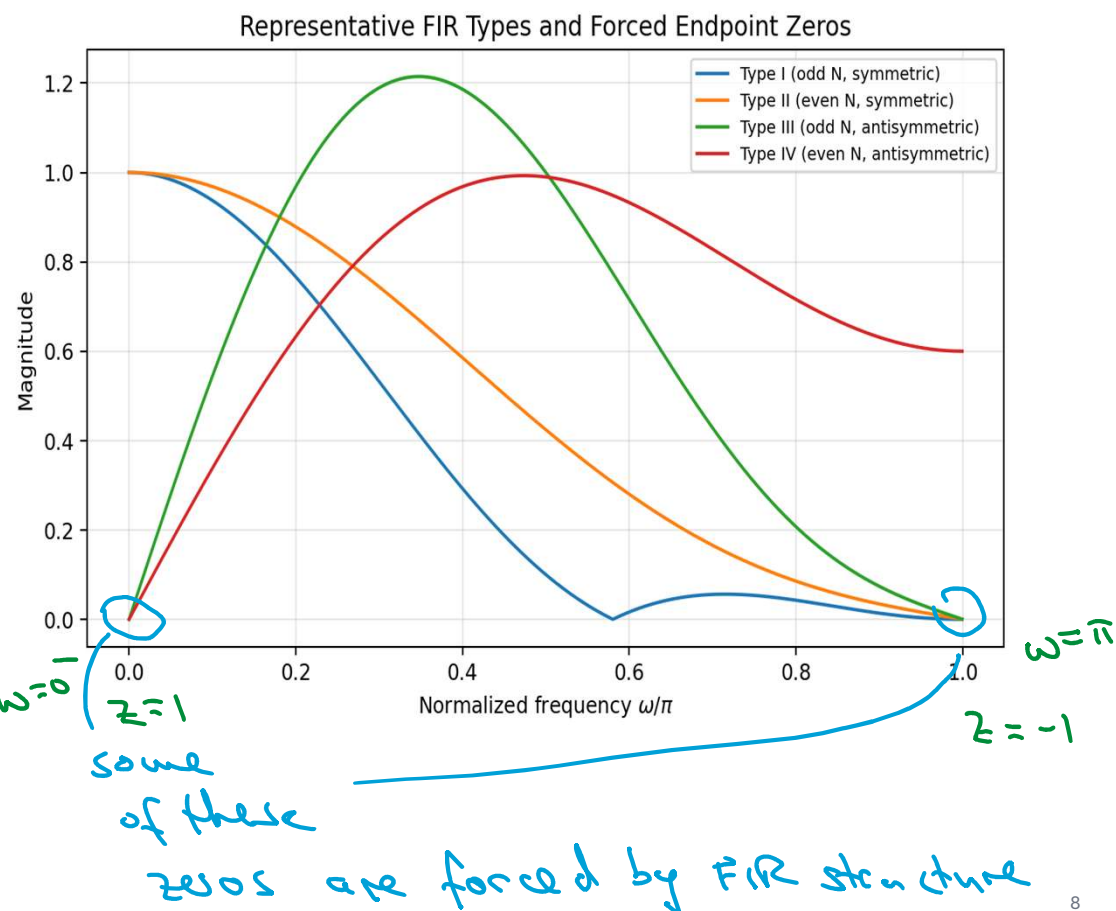
The four real FIR linear-phase types

- Type I • odd N, symmetric. *even M*
- Type II • even N, symmetric. *odd M*
- Type III • odd N, antisymmetric. *even M*
- Type IV • even N, antisymmetric. *odd M*

N = M + 1
of taps → *order of filter*

Why four?

Parity of length and choice of symmetry determine whether the midpoint is a sample or half-sample and which endpoint zeros are forced.



Structural Constraints

$h[n]$ real-valued

$$H(z) = \sum_{n=0}^M h[n] z^{-n} \quad \text{if } \underline{h[n] = h[M-n]} \quad \text{then}$$

if $h[n] = -h[M-n]$
antisymmetric
 $\Rightarrow H(z) = -z^{-M} H(z^{-1})$

$$H(z) = \sum_{n=0}^M h[M-n] z^{-n} = \sum_{k=M}^0 h[k] z^k z^{-M}$$

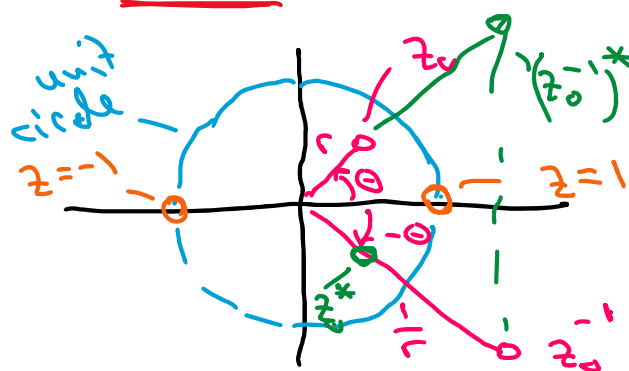
$M-k \leftarrow n \quad k \leftarrow M-n$

$$= z^{-M} \cdot H(z^{-1})$$

$$\Rightarrow \underline{H(z) = z^{-M} \cdot H(z^{-1})}$$

Antisymmetry: $H(z) = -z^{-M} H(z^{-1})$

If $z_0 = re^{j\theta}$ is a zero of $H(z) \Rightarrow H(z_0) = (z_0)^{-M} H(z_0^{-1}) = 0$



$z_0^*, (z_0^{-1})^*$

special constraints

$\Rightarrow \underline{z_0^{-1} = \frac{1}{r} e^{-j\theta} = \frac{1}{r} e^{-j\theta}}$ is also a zero
for real-valued $h[n]$
we need complex conjugates

z_0, z_0^{-1} are in reciprocal locations

Type I: odd length, symmetric

$$N \text{ odd}, \quad h[n] = h[M - n]$$

M even

symmetric

- Group delay is integer: $M/2 = (N-1)/2$.
- No zero at $z=1$ or $z=-1$ is forced by symmetry alone.
- Can realize low-pass, high-pass, band-pass, band-stop, and many other real linear-phase responses.

M: filter order

N = M + 1 is the number of filter taps

Type II: even length, symmetric

$$N_{\text{even}}, \quad h[n] = h[M - n]$$

M odd

symmetric

$$H(e^{j\pi}) = 0 \quad \Longleftrightarrow \quad z = -1 \text{ is a zero}$$

- Group delay is half-integer.
- Because gain must vanish at Nyquist, a Type II filter cannot realize a high-pass response with nonzero $H(e^{j\pi})$.

Type III: odd length, antisymmetric

$$N \text{ odd, } h[n] = -h[M - n]$$

M even *antisymmetric*

$$H(e^{j0}) = 0, \quad H(e^{j\pi}) = 0$$

- Forced zeros at $z=1$ and $z=-1$.
- Cannot have nonzero DC or Nyquist gain.
- Useful for differentiator-like or Hilbert-transform structures over an interior band.

Type IV: even length, antisymmetric

$$N_{\text{even}}, \quad h[n] = -h[M - n]$$

M odd

antisymmetric

$$H(e^{j0}) = 0$$

- Forced zero at DC, but not necessarily at Nyquist.
- Half-integer group delay.
- Well suited to Hilbert transformers and differentiators, where zero DC response is natural.

Concept check: why Type II cannot be high-pass

- A high-pass normally requires appreciable gain near $\omega=\pi$.
- Type II symmetry forces $H(e^{j\pi})=0$.
- Therefore the required endpoint behavior contradicts the symmetry constraint.
- Strong answer: this is not a matter of filter order; it is structural and holds for every Type II FIR.

$$H(z) = \sum_{n=0}^M h[n] z^{-n}$$

$$H(e^{j\pi}) = H(-1) = h[0] - h[1] + h[2] - h[3] \dots - h[M]$$

M is odd

$$h[n] = h[M-n] \Rightarrow H(-1) = 0$$

Zero symmetry for real linear-phase FIR filters

$$z_0 = re^{j\theta} \quad \frac{1}{z_0} = \frac{1}{r} e^{-j\theta}$$

structural constraint

$$\underline{z_0}, \quad \underline{z_0^*}, \quad \underline{\frac{1}{z_0}}, \quad \underline{\frac{1}{z_0^*}}$$

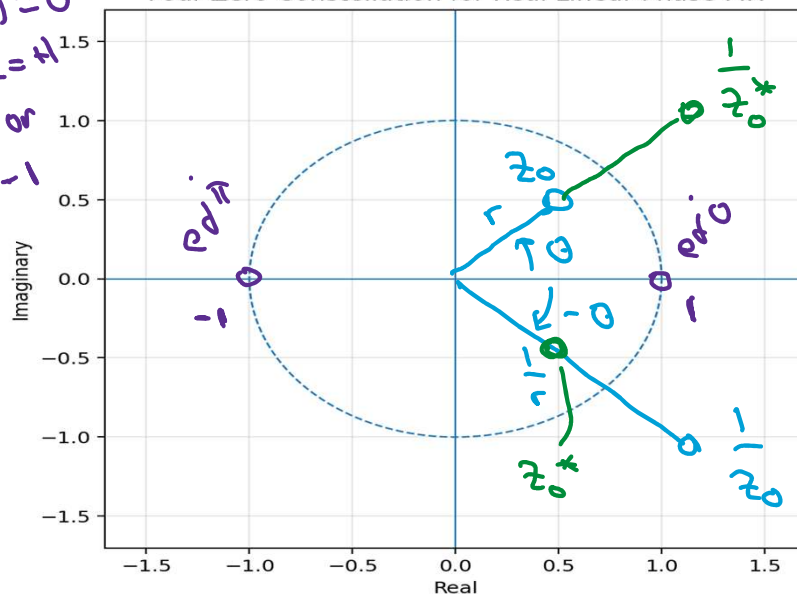
$H(z) = \pm z^{-M} \cdot H(z^{-1})$ may force $H(z) = 0$ at $z = 1$ or $z = -1$

Why four?

Real coefficients give conjugate symmetry; palindromic / anti-palindromic linear-phase structure gives reciprocal symmetry.

at $z = 1$ or $z = -1$

Four-Zero Constellation for Real Linear-Phase FIR



Degenerate cases

Zeros on the unit circle or real axis can collapse the four-point set into pairs or individual points.

to get real valued $h[n]$ we also need z_0^* and $\frac{1}{z_0^*}$

$H(z) = z^{-M} H(z^{-1})$
 $\Rightarrow$ if z_0 is zero of $H(z)$ then z_0^{-1} must be zero of $H(z)$

from symmetry constraint

Polynomial viewpoint

- A symmetric FIR has a palindromic coefficient polynomial; an antisymmetric FIR has an anti-palindromic one.
- If z_0 is a zero, reciprocal symmetry forces $1/z_0^*$ (with conjugation as required for real coefficients).
- This zero geometry is the z -plane fingerprint of linear phase.

+ is for symmetric FIR coeff $h[n] = h[M-n]$

$$H(z) = \pm z^{-M} H(z^{-1})$$

- is for antisymmetric FIR coeff $h[n] = -h[M-n]$

$h[n]$ are real-valued

Generalized phase: sign changes create apparent phase jumps

- When the real amplitude factor $A(\omega)$ crosses zero, its sign changes.
- A negative real amplitude contributes an additional π to the displayed phase.
- The unwrapped phase may show jumps, but the underlying delay term remains $-\omega M/2$.
- Use group delay away from magnitude zeros to diagnose the physical timing behavior.

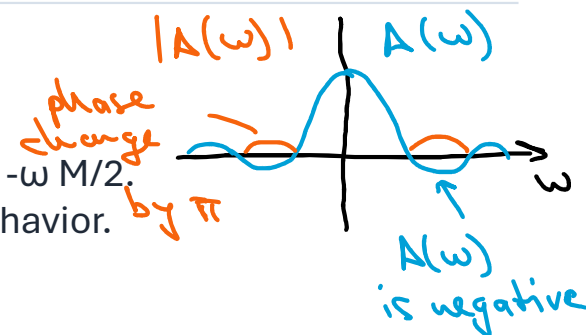
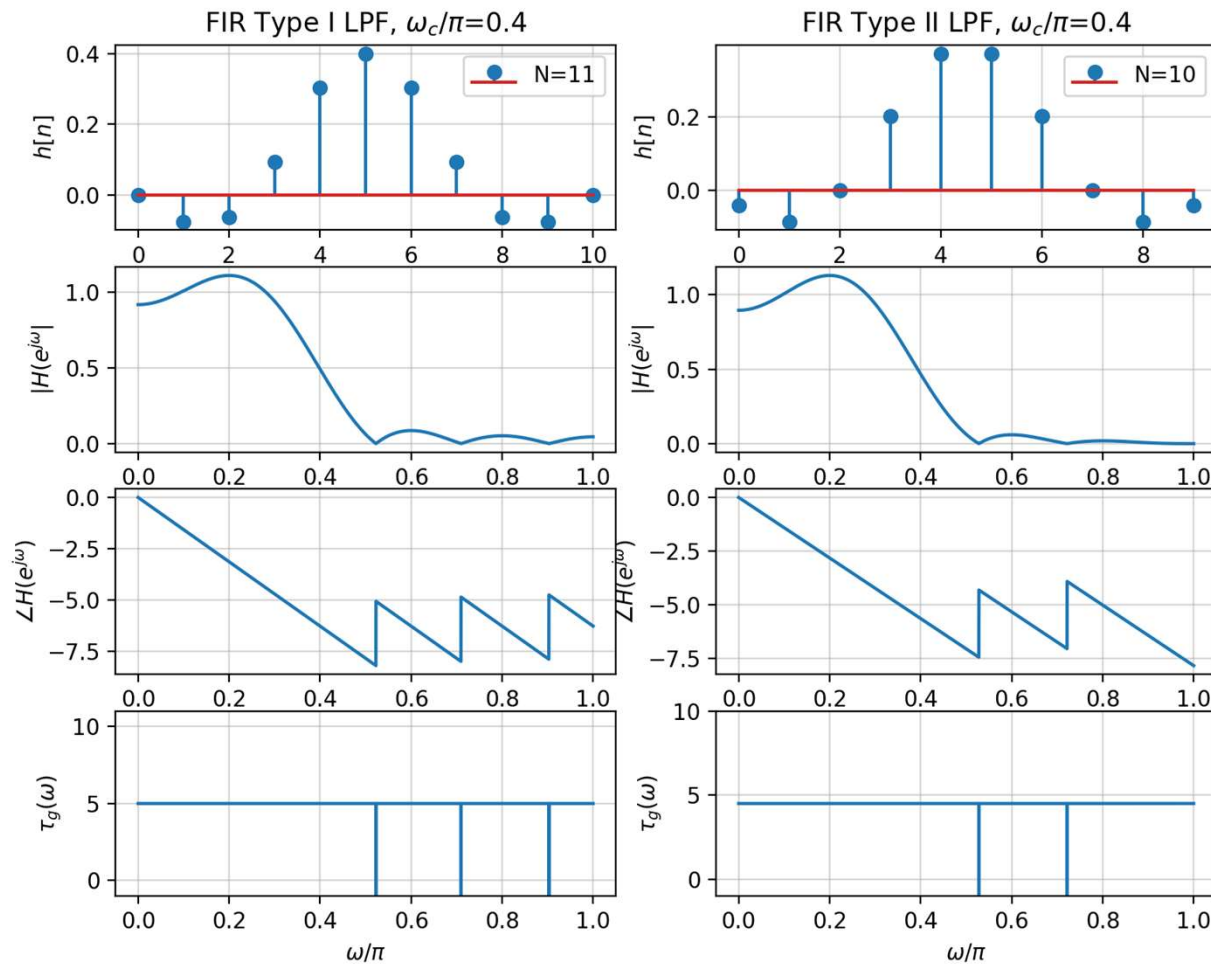


Table of constraints

Type	M	Symmetry	Forced zero at $\omega=0$	Forced zero at $\omega=\pi$	
I	even	symmetric	no	no	LPF, HPF, BPF, BSF
II	odd	symmetric	no	yes	no HPF, BSF
III	even	anti-symmetric	yes	yes	just BPFs
IV	odd	anti-symmetric	yes	no	no LPF, BSF

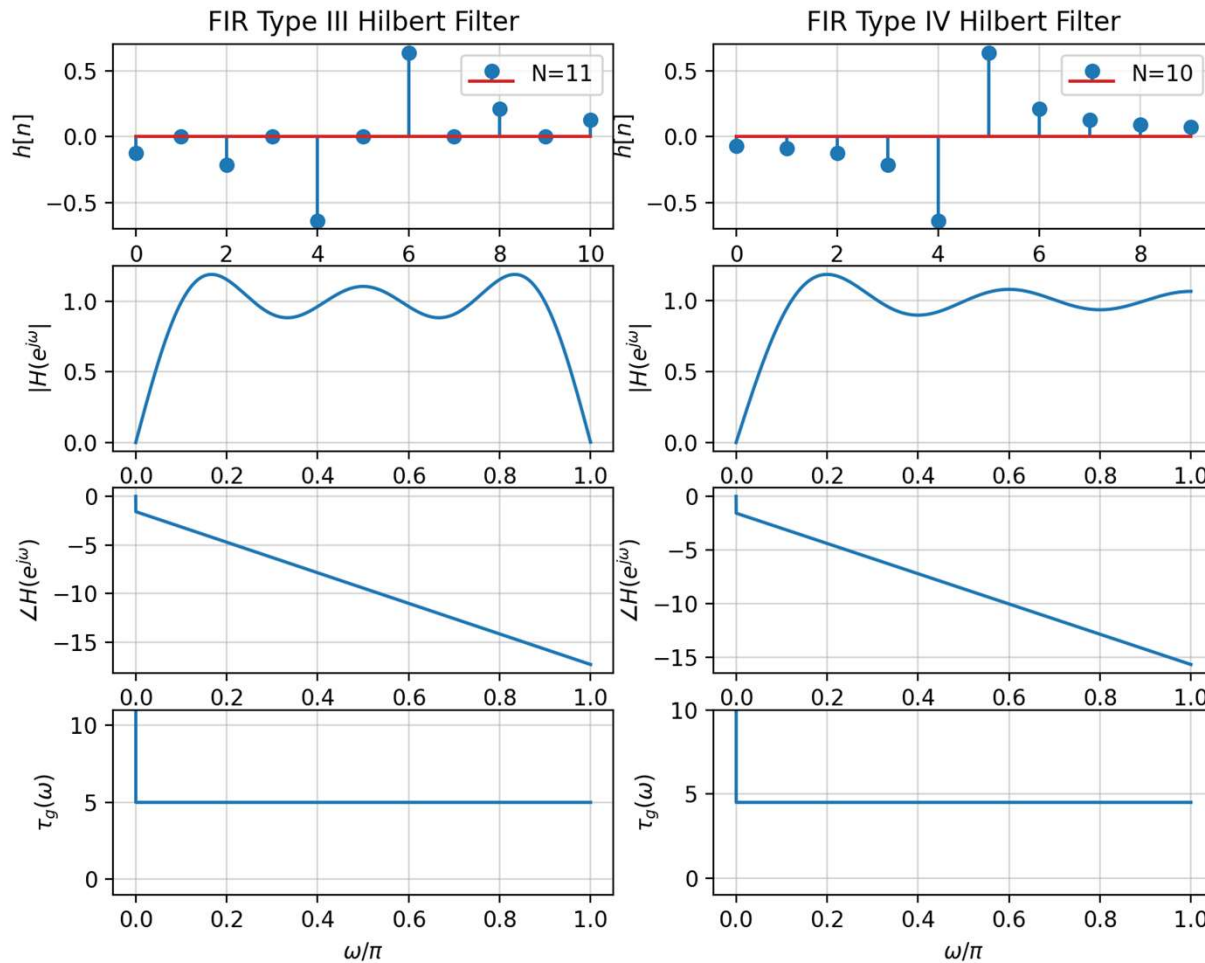
Type I and II FIR Filter Examples



The Type II filter has a forced zero at $z = -1$

Note the phase jumps by π when $H(e^{j\omega})$ goes negative (and is displayed as $|H(e^{j\omega})|$)

Type III and IV FIR Filter Examples



The Type III filter has forced zeros at $z = \pm 1$ and the Type IV filter has a forced zero at $z = 1$.

Note the phase of $\beta = -\pi/2$ at $\omega = 0^+$

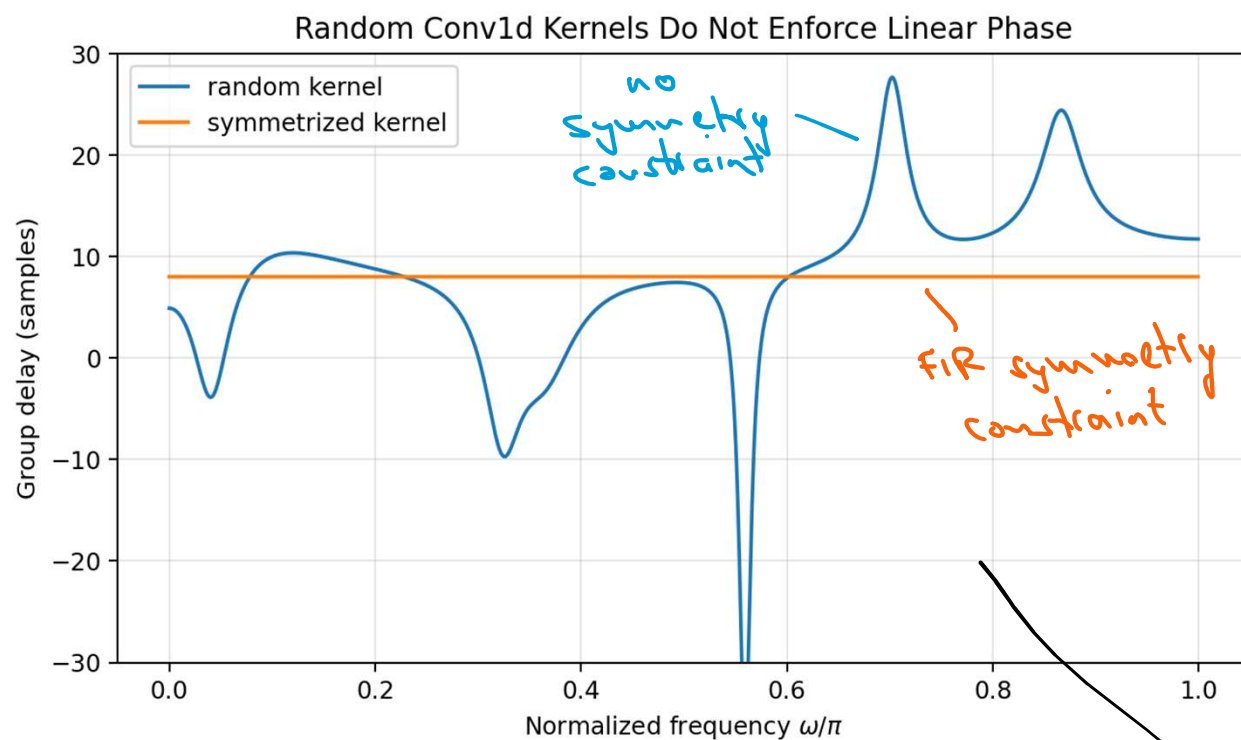
Conv1d is a trainable FIR-like local operator

$$y_c[n] = b_c + \sum_m \sum_{k=0}^{K-1} w_{c,m}[k] x_m[n+k]$$

w: trainable weights
corresponding to impulse response $h[n]$ in traditional DSP
cross-correlation

- For one input/output channel, the kernel is simply a finite set of tap weights.
- Multiple channels generalize one FIR to a bank of learned finite-support cross-correlations.
- Deep-learning libraries use correlation ordering, but the FIR interpretation remains useful.

A random Conv1d kernel does not have linear phase by default



Read the plot

A generic random kernel has no symmetry constraint, so its group delay varies. Symmetrizing the coefficients forces the FIR into a linear-phase family.

Group delay

How to impose symmetry in a trainable kernel

$$h = [a_0, a_1, \dots, a_R, \dots, a_1, a_0]$$

- Learn only one half of the taps (and the center tap for odd length).
- Construct the other half by reversal inside the forward pass.
- Autograd still trains the independent parameters, but every realized kernel remains symmetric.
- This is an architectural inductive bias, not a post-hoc correction.

↑ substitute this part using symmetry

let ML train this part

PyTorch symmetry parameterization

```
half = nn.Parameter(torch.randn(R))
center = nn.Parameter(torch.randn(1))

def kernel(self):
    return torch.cat([
        self.half,
        self.center,
        torch.flip(self.half, dims=[0])
    ])

# Use the constructed kernel in conv1d.
# Symmetry is exact at every optimizer step.
```

concatenate

for type I FIR

What inductive bias are we adding?

- Benefit: exact generalized linear phase and fewer independent parameters.
- Benefit: preserves relative timing when that invariance is known to matter.
- Cost: the model cannot learn an arbitrary nonlinear-phase kernel even if the task would benefit from one.
- Engineering principle: constrain only what physics or deployment requirements justify.

Fixed, constrained, or fully learned front end?

Fixed DSP

Design taps from a frequency specification; no learning.

Constrained learned

Learn taps but enforce symmetry / band structure / sparsity.

Unconstrained learned

Let the task loss select arbitrary finite kernels.

- These are not competing religions; they are different locations on the bias–flexibility spectrum.

Multirate transition: what linear phase does and does not guarantee

- Anti-alias filtering requires adequate stopband attenuation before downsampling.
- Linear phase is not mathematically required to suppress aliases.
- However, linear-phase FIRs are often preferred when relative waveform timing and alignment must be preserved across sample-rate conversion.
- Next week: polyphase / multirate structure makes FIR implementation especially attractive.

Concept check

- A 16-tap symmetric FIR is which type?
- What endpoint must be zero?
- Can it be a conventional high-pass with nonzero Nyquist gain?

Concept check

- A 16-tap symmetric FIR is which type?
- Answer: Type II because N is even and the taps are symmetric.
- What endpoint must be zero? $H(e^{j\pi})=0$.
- Can it be a conventional high-pass with nonzero Nyquist gain? No.

Python demo: random versus symmetry-constrained kernels

```
rng = np.random.default_rng(4)
h_rand = rng.normal(size=17)
h_sym = 0.5*(h_rand + h_rand[::-1])

_, gd_rand = signal.group_delay((h_rand, [1.0]), w=2048)
_, gd_sym = signal.group_delay((h_sym, [1.0]), w=2048)

# PyTorch fixed FIR equivalence
kt = torch.tensor(h[::-1].copy()).view(1,1,-1)
yt = F.conv1d(F.pad(xt, (len(h)-1,0)), kt)
```

Lecture 10 synthesis

- Generalized linear phase means a constant delay plus a real amplitude factor.
- Symmetry / antisymmetry creates exactly four real FIR types with structural endpoint constraints.
- Linear-phase zero patterns appear in conjugate-reciprocal constellations.
- Conv1d kernels are not linear phase unless the architecture or parameterization enforces it.